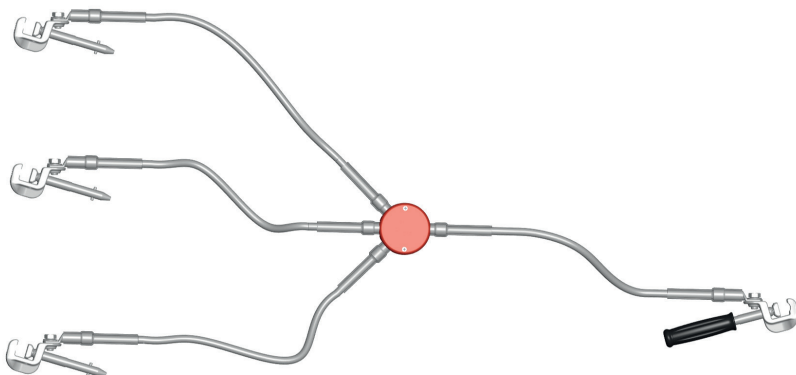


PFISTERER

Earthing and short-circuiting device
040 106 020

OPERATING INSTRUCTIONS

Earthing and short-circuiting device



THE POWER CONNECTION

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1 General information

Earthing and short-circuiting devices (as per EN/IEC 61230:2009) are hand-operated devices that are brought carefully up to the connection points of electrical system components and connected for the purpose of earthing and short-circuiting.

1.1 Information about the operating instructions

These operating instructions describe the safe and proper use of the described product. The safety information and instructions must be observed in addition to the locally applicable accident prevention regulations relating to the area of application of the product and general safety rules.

Carefully read through the operating instructions, especially the section on safety and the relevant safety information, before starting any work on the product. It is vital that what is read has also been understood. The operating instructions are a constituent part of the product. They must be kept accessible at all times in the immediate vicinity of the product (control room, etc.).

1.2 Operating instructions contents

Every person who is commissioned to perform work on or with the product must have read and understood the operating instructions before working on/with the product. This requirement applies even if the person has already worked with such a product or a similar product or has been trained by the manufacturer. Knowledge of the contents of the operating instructions is one of the requirements for protecting personnel from hazards as well as for preventing errors, thus ensuring the product is used in a safe and trouble-free manner. The operating company is advised to have the staff confirm that they have familiarized themselves with the contents of these operating instructions.

1.3 Explanation of symbols, warnings

Warnings are indicated by a triangular safety symbol and a signal word. The signal word describes the severity of the danger that could occur, with the following meanings:



DANGER

Indicates an immediate danger with high risk that could result in death or severe bodily injuries if not avoided.



WARNING

Indicates a possible danger with moderate risk that could result in death or severe bodily injuries if not avoided.

**CAUTION**

Indicates a danger with low risk that could result in light or moderate bodily injuries or property damage if not avoided.

1.4 Other symbols

**IMPORTANT**

Indicates a required behaviour or an activity required for proper handling of the product.
Failure to observe these instructions could result in property damage or product malfunctions.

**NOTE**

Indicates usage tips and instructions that help get the most out of the product.

1.5 Copyright protection

The operating instructions are confidential. They are intended solely for persons who work on and with the product.

All contents, text, drawings, images and other representations are protected under copyright law and are subject to other intellectual property rights. Any misuse is punishable by law.

Disclosure to third parties, including making copies of any type or form - even of extracts - as well as the use and/or communication of the contents without written permission from the manufacturer is prohibited. Any infringements shall be subject to claims for compensation. We reserve the right to assert further claims.

We reserve the right to exercise any of our intellectual property rights.

1.6 Liability and warranty

All information and data contained in these operating instructions were compiled under consideration of the applicable regulations, the technological state of the art and our many years of expertise and experience.

The operating instructions are to be read carefully before starting work on and with the product. The manufacturer shall accept no liability for any damage or malfunctions resulting from failure to observe the operating instructions.

The text and images contained in the instructions do not necessarily correspond to the scope of delivery. The illustrations are not to scale. Due to the fact that special versions or additional options may have been ordered or innovative technical modifications may have been made, the actual scope of delivery can differ from the explanations and illustrations described in these instructions. Please direct any queries to the manufacturer.

We reserve the right to make technical modifications to the product within the framework of improving performance characteristics and further development.

1.7 Warranty

The terms and conditions of warranty are defined in the purchase agreement and the general terms and conditions of the manufacturer.

On return of defective parts or inspection on site, the manufacturer shall ultimately make a decision on the validity of a warranty claim.

1.8 Spare parts

Only use genuine spare parts of the manufacturer.

The use of unapproved spare parts voids all claims concerning service, compensation for damages and liability against the manufacturer or its agents, dealers or representatives.

2 Safety

At the time of its development and manufacture, the product was constructed according to currently applicable, accepted codes of engineering practice and is safe to operate. Nevertheless, the product can pose dangers if it is not used by professionally trained personnel or is used improperly or in a manner deemed not for its intended purpose.

2.1 Intended use

The earthing and short-circuiting device is designed exclusively as a safety component for temporary installation in a system.

It serves to divert residual voltages and short-circuit currents.

The earthing and short-circuiting device may only be used on system components for which it is rated, taking into consideration the short-circuit currents and short-circuit times that arise.

Any other than the intended use of the product is prohibited and is deemed as not for its intended purpose.

Any claims against the manufacturer and/or his authorized representatives for damage resulting from use of the product not for its intended purpose shall not be accepted.

The operator shall be solely liable for any damage resulting from use of the product other than its intended purpose.

Operational reliability/safety can only be ensured if the product is used as intended.

It may only be used in accordance with these operating instructions.

Earthing and short-circuiting devices that have been subjected to a short-circuit current or that are even slightly suspected of being unsafe must be excluded from further use.

2.2 Foreseeable improper use

Use of the product outside of the bounds of the purpose specified by PFISTERER is considered to be unintended use and therefore improper use.

Such improper uses include:

- If the product is used for transmitting current.
- If short-circuiting cables are connected without an earthing stick.
- Use of short-circuiting cables that have already been subjected to a short-circuit current.
- Use of short-circuiting cables with breaks.
- Use of short-circuiting cables with damaged insulation.
- Use of short-circuiting cables when the cable or the connection terminal exhibits signs of oxidation.

2.3 Changes and modifications to the product

To avoid hazards and to ensure optimum performance, no changes, additions or modifications may be made to the product that have not been expressly approved by the manufacturer.



NOTE

For device identification and regular inspections, it is permitted to affix labels at certain points on the product.

2.4 Hazards posed by the product

The state-of-the-art design and configuration of the product correspond to the latest technological developments.

The product is safe when used for its intended purpose. Nevertheless, a certain degree of residual risk remains!



DANGER

Danger - electrical current!

Electrical power can cause very serious injury. Damage to the insulation or individual components may result in the danger of death by electrical current.

2.5 Work safety

Personal injury and material damage while working with and on the product can be avoided by following the safety precautions and instructions specified in these operating instructions.

Failure to comply with these instructions can result in personal injury and damage to or destruction of the product.

Any liability and compensation claims against the manufacturer or his authorized representative resulting from non-compliance with the safety precautions and instructions specified in this operating instructions as well as the accident prevention regulations applicable to the area of application of the product and the general safety regulations shall not be accepted.

2.6 Responsibility of the operating company

The product may only be used in a technically perfect and operationally reliable condition. The information given in the operating instructions must be followed in its entirety and without restrictions.

In addition to the safety precautions and instructions specified in these operating instructions, the locally applicable accident prevention regulations relating to the area of use of the product and general safety regulations as well as the applicable environmental protection regulations must be observed and followed.

The operating company and personnel authorized by the operating company shall be responsible for fault-free operation of the product and for clear definitions of the responsibilities for operation and repair of the product.

2.7 Requirements for the personnel

Technical personnel are persons who, on the basis of their technical training, knowledge and experience as well as knowledge of the applicable regulations, can assess the tasks assigned to them and recognize potential hazards.

The local age and job-specific regulations applicable at the place of use of the product must be taken into account with regard to personnel selection. Such specialists or trained persons are defined in EN 50110-1. The earthing and short-circuiting device complies with EN/IEC 61230.

The responsibilities for working on and with the product (operation, repair) must be clearly defined and complied with to avoid any ambiguous division of responsibilities in terms of safety.

Only persons who can be trusted to work reliably are permitted to work on and with the product. All working methods that impair the safety of persons, the environment or the product are prohibited.

Persons under the influence of drugs, alcohol or medication that impedes their reactions must not work on or with the product.

The personnel must immediately report to the operator any changes to the product that impair safety and prevent further use of the product.

2.8 Personal Protective Equipment (PPE)

The requirements of the system owner must be complied with. When working on and with the product, it is fundamentally recommended that the following be worn:



Protective clothing, close-fitting working clothes (low tear resistance, close-fitting sleeves, no rings or other jewellery etc.).



Safety footwear to provide protection from heavy, falling objects and slipping on slippery floors.



Safety helmet to provide protection from falling and flung objects and materials.



Face shield to protect the eyes and face against flame, sparks, embers, hot particles and exhaust gases.



Safety goggles to protect the eyes from flash burns.



Safety gloves to protect the hands from friction, scrapes, pricks or deeper injuries as well as from hot surfaces.



Wear a bright, reflective vest or clothing to be more visible.



Use a safety harness to protect against falls when working above ground level.

The following situations in particular require a safety harness:

- Work on high-voltage pylons or power poles.
- All work above ground level.

2.9 Special dangers



WARNING

Risk of injury due to uncontrolled sudden movement of the device!

In the event of a short-circuit, the device may be subjected to strong mechanical forces, resulting in uncontrolled movement that can cause severe injuries.

- Wear face protection during all work.
- Wear a safety harness during work above ground level.
- Always install the device using earthing sticks.



WARNING

Risk of injury due to electric arcing!

In the event of a short-circuit, eyes can suffer flash burns from electric arcing that produces intensive visible infra-red and ultraviolet radiation.

- Wear safety goggles during all work.



WARNING

Risk of injury due to corrosive vapours!

In the event of a short circuit, high temperatures release corrosive vapours that can cause severe injuries of the respiratory tract, eyes and skin.

- Wear face protection and personal protective equipment during all work.
- Immediately leave the danger area.
- Always work in pairs.



CAUTION

Risk of burns due to hot surfaces!

Contact with hot components can cause burns.

- Wear protective clothing and gloves during all work in the vicinity of hot components.
- Allow the earthing and short-circuiting device to cool after use in which current was diverted.

3 Transportation / storage



IMPORTANT

The earthing and short-circuiting device, including earthing stick, must be handled with great care and protected appropriately to prevent soiling and damage.

III

3.1 Scope of supply

Check the scope of delivery for completeness. Contact the manufacturer immediately if parts are missing. Reference is made to the terms and conditions of purchase and delivery.

On delivery of the product, any damage as a result of defective packaging or that has been incurred during transportation must be reported immediately to the shipping company, the insurer and the supplier. Ensure that damage is minimised and that further damage is avoided.

3.2 Packaging

The main packaging materials are wood, cardboard and plastic (film, foam). If no corresponding agreement has been made with regard to its return, the packaging material will remain with the customer.

Environmentally acceptable disposal in accordance with the applicable waste disposal regulations must be ensured. If necessary have a waste disposal company dispose of the packaging material.

3.3 Transport

Ensure that the earthing and short-circuiting device remains dry during transport. Use an appropriate container for transport.

3.4 Storage

The storage period is generally not limited. The product must be stored under the following storage conditions:

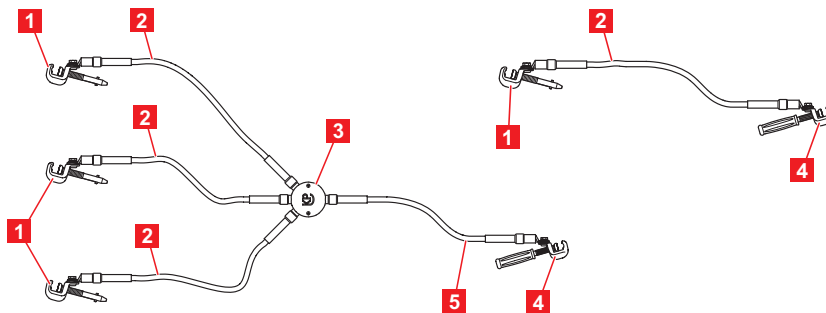
- Store in dry place. Ensure that the packed items are not stored outdoors. Additional steps must be taken to ensure that the floor of the storage room is dry during the storage period.
- Protect from direct sunlight.
- Store free of dust.
- Avoid mechanical vibrations and damage.

4 Product description

4.1 Function description

Earthing and short-circuiting devices are safety devices that draw off the residual voltage in disconnected systems and divert any short-circuit current to earth potential in the event of unintended reactivation.

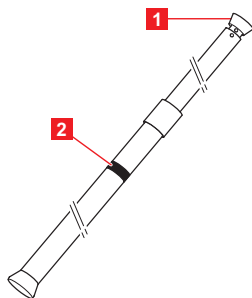
4.2 Overview



- | | |
|-----------------------------|-----------------------------|
| 1 Phase connection terminal | 4 Earth connection terminal |
| 2 Short-circuiting cables | 5 Earthing cable |
| 3 Nodal point housing | |

Special variants may differ from what is depicted here.

4.3 Accessories (optional)



Earthing stick

The earthing stick is used to bring the phase connection terminals to the system and connect them.

The black ring (2) indicates the safe gripping area on the handle.

5 Technical data



NOTE

Information on technical data can also be found on the product sticker on the packaging.

5.1 Connected loads Rated current for copper cables

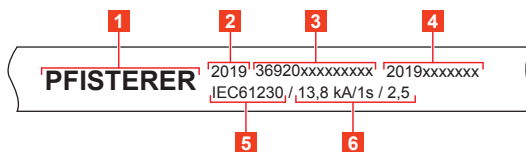


NOTE

Information on the cable cross section can be found on every cable.

Cable cross section	Max. short-circuit current	Max. short-circuit current
mm ²	$I_{sc} = <0.5 \text{ s}$	$I_{sc} = 1.0 \text{ s}$
25	7,000	4,900
35	10,000	7,000
50	14,000	9,900
70	19,500	13,800
95	26,500	18,700
120	33,500	23,700
150	42,000	29,600

Rating information can be found on the labels on the short-circuiting cables.



- | | |
|--------------------------|----------------------------------|
| 1 Manufacturer | 4 Specific project number |
| 2 Production year | 5 Manufacturing standard |
| 3 Article number | 6 Rated values |

5.2 Operating conditions

Description	Value	Unit
Ambient temperature range	-25 - +55	°C
Max. relative humidity	20 - 96	%

6 Assembly



DANGER

Risk of death due to adjacent electrically live parts!

If adjacent system components that are still live cannot be switched off, there is a risk of fatal injury if contact is made with these system components.

- Isolate adjacent live system components with firmly and securely affixed covers.
- Cordon off the system components at a good distance.
- Maintain all safety measures until the work has been fully completed and the earthing and short-circuiting device has been fully disconnected.



DANGER

Danger - electrical current!

The earthing and short-circuiting device must always be connected first to the earthing terminal at earth potential.



NOTE

If earthing and short-circuiting devices are connected in parallel to achieve specific total cable cross sections, the following conditions must be met.

- Use cables with the same lengths and cross sections.
- Use the same connecting points and connecting parts.
- Install the earthing and short-circuiting devices closely together with cables running in parallel.
- Reduce the permissible electrical load to 75% of the rating for the cable cross section.

VI

6.1 Inspection before use

For safety reasons, the earthing and short-circuiting devices must be handled carefully and given a thorough visual inspection before every use.



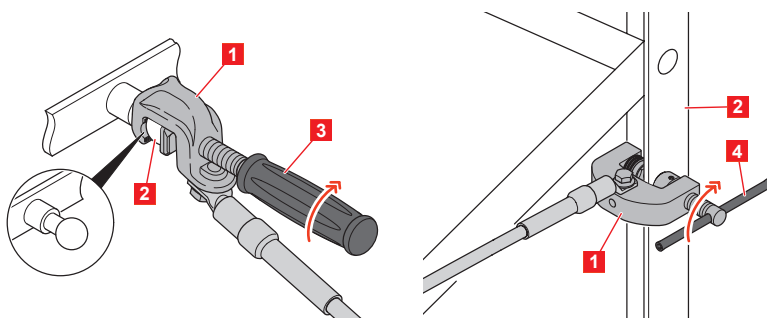
IMPORTANT

In the event of visible damage to connecting parts, connections and cable sheathing or if a bare conductor is exposed, the earthing and short-circuiting device must be excluded from use.

- 1 Check the contact surfaces of the connecting parts before use. Remove any deposits and oxidation traces with a brush.
- 2 Check based on the rated values and technical data whether the earthing and short-circuiting device is suitable for the intended use.
- 3 Check connection terminals for a firm seat and proper function. The tightening torque of the screw connection is 56 Nm.
- 4 Check that the earthing stick is suitable and functioning properly.

6.2 Installation of earth connection terminal

- 1 Attach or contact the earth connection terminal (1) at the earthing point (2). Contact surfaces must rest completely against the bare earthing point.
- 2 Tightly screw on the earth connection terminal (1) by hand using the handle (3) or screw wings (4).



6.3 Installation of phase connection terminal



DANGER Danger - electrical current!

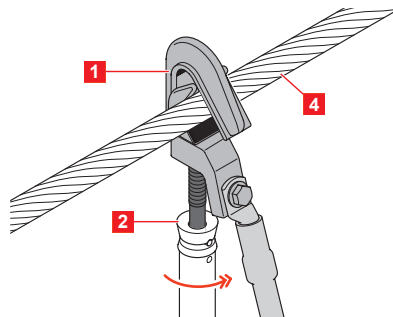
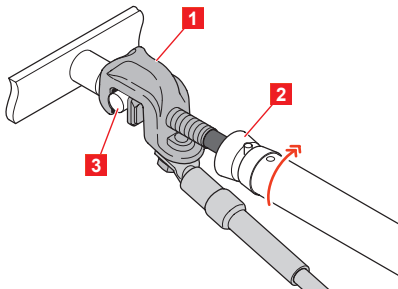
If the earth connection terminal is not connected to the earth potential first, there is a risk of fatal injury due to electrical current.

- Always install the earth connection terminal first.



NOTE
The earthing stick is not included in the delivery. Observe the operating instructions of the earthing stick before use.

- 1 Slide the spindle of the phase connection terminal (1) onto the earthing stick (2) and lock it in place.
- 2 Use the earthing stick (2) to connect the phase connection terminal (1) to the phase connection point (3) or the stranded conductor (4).
- 3 Firmly screw the phase connection terminal to the earthing stick (2).

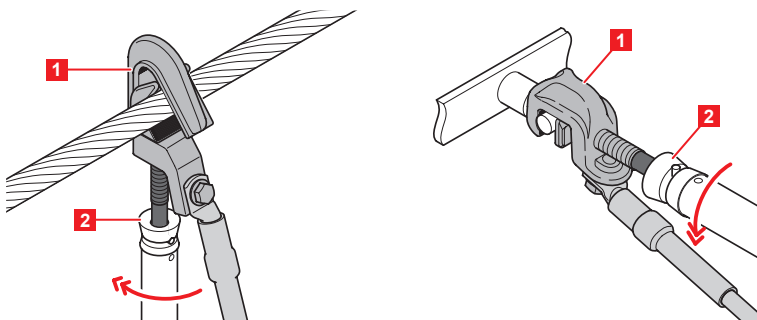


- 4 Remove earthing stick.
- 5 Repeat steps 1 to 4 to connect further phase connection terminals.

VI

7 Disassembly

- 1 Slide the earthing stick (2) onto the spindle of the phase connection terminal (1) and lock it in place.
- 2 Turn the earthing stick until the phase connection terminal is released.
- 3 Use the earthing stick to remove the phase connection terminal from the phase connection point or stranded conductor.



- 4 Repeat steps 1 to 3 to remove all other phase connection terminals.

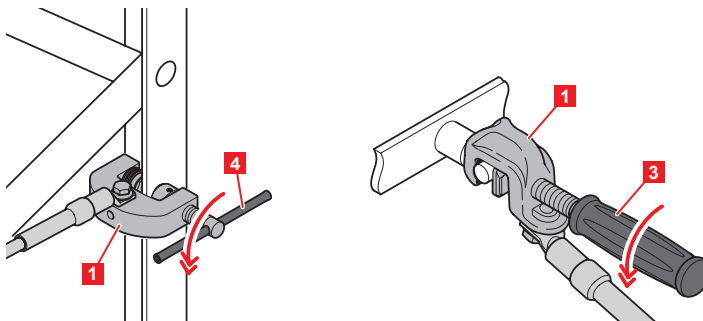


DANGER
Risk of death due to electrical current!

If not all phase connection terminals are removed, there is a risk of death due to electrical current.

- First remove all phase connection terminals before removing the earth connection terminal.

- 5 Completely remove the earth connection terminal (1) by hand using the handle (3) or screw wings (4).
- 6 Disconnect the earth connection terminal (1) at the earthing point.



8 Maintenance



DANGER **Risk of death due to electrical current!**

Contact with live components can result in fatalities.

- All maintenance and cleaning work may only be performed while the device is disconnected and in a dry environment.

8.1 Cleaning

- Check the device for soiling after every use.
- All components are to be cleaned with a soft, damp cloth.
- Do not use any aggressive cleaning agents.
- Make sure that the device is completely dry before it is used again.

8.2 Maintenance plan

- The operating company must have the necessary maintenance work carried out immediately if increased wear on the individual components or assemblies is identified during routine inspections.

Interval	Maintenance work	Performed by
After use	Clean device	Qualified electrician
	Apply a thin coat of acid-free grease to all threads	Qualified electrician
As needed	Replacement of damaged terminals	Manufacturer, master electrician
	Replacement of damaged cables	Manufacturer, master electrician

VIII

8.3 Testing

- To verify the proper condition of the earthing and short-circuiting device, the operating company must develop its own guidelines in consideration of the technical rules for operating safety and EN/IEC 61230:2009 Annex C and define the regular inspection interval.
- Inspection intervals must be chosen such that defects arising during use can be identified in a timely fashion.

8.4 Spare parts

- Repairs to earthing and short-circuiting devices may only be performed with components approved by the manufacturer.
- Short-circuiting cables can be replaced individually.
- When replacing connection terminals on earthing and short-circuiting cables, ensure that a tightening torque of 56 Nm is applied.

9 Disposal

If no return or disposal agreement has been made, consign disassembled components after proper disassembly for recycling as follows:

- Scrap the metallic materials.
- Send plastic elements to plastic recycling.
- Sort and dispose of remaining components according to material type.

**IMPORTANT**

Danger of environmental damage!
Lubricants and other additives are subject to special waste treatment and may only be disposed of by specialist waste disposal companies.

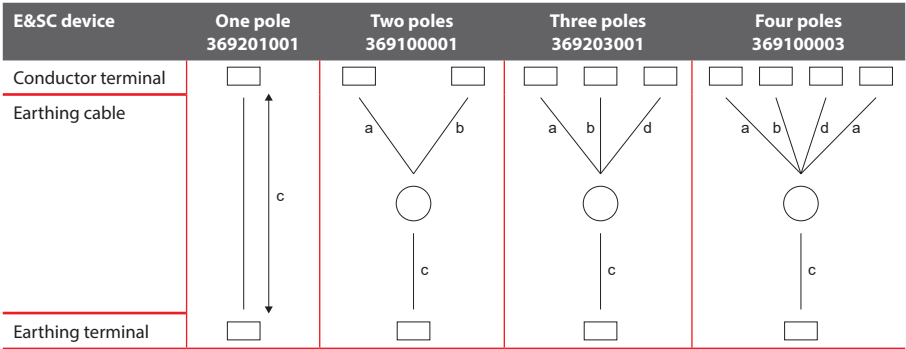
Earthing and short-circuiting device inquiry

Dear customer,

To process your inquiry quickly and save you any inconvenience, we ask that you please provide the information requested below.

Company:		Name:	
Telephone:		Date:	
Email:		Signature:	

1



2

Cross section		Length	
Short-circuiting cable:		Short-circuiting cable a:	
Cross section		Length	
Earthing cable:		Short-circuiting cable b:	
Conductor terminal:		Length	
		Short-circuiting cable d:	
Earthing terminal:		Length	
		Earthing cable c:	

3

Information about PFISTERER terminals can be obtained by telephone, by email or in the online catalogue on our website.

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